



Energy Efficiency Glazing Provisions for BCA Volume Two



Changes Introduced in BCA 2010
and
Advanced Glazing Calculator Techniques

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Introduction

Energy efficiency provisions, including elemental Deemed-to-Satisfy (DtS) Provisions for residential glazing, have been in BCA Volume Two since 2003. The ABCB provides a range of glazing calculators that are based on the glazing provisions in BCA Part 3.12.2. Although it is possible to work directly with the formulae in those provisions, the calculators guide you through the information required and provide some checks on the information entered.

The stringency of the BCA Volume Two glazing provisions was increased in BCA 2010 as part of the move to a nominal 6-star equivalent for the elemental DtS Provisions, and a new glazing calculation methodology was applied for buildings in climate zones 2 to 8. The combined effect of these changes is that it may be challenging to design a complying glazing solution using the elemental DtS Provisions and associated glazing calculator.

This document explains the changes to the glazing provisions and the glazing calculator, and describes some advanced ways to use the glazing calculator. The information is organised in the following sections:

1. Glazing Basics
2. Changes in Calculation Methodology for Climate Zones 2 to 8
3. Changes to the Glazing Calculator Layout
4. Advanced Glazing Calculator Techniques
5. Glazing in Alterations and Additions

Where compliance with the elemental DtS Provisions for glazing seems too demanding, the whole-of-house energy rating approach may prove to be more flexible. It allows the trading of performance between building elements and could make simpler glazing arrangements viable.

Early resolution of the glazing design, including the size, orientation and shading of glazing elements, is critical as high-performance glazing alone may no longer be relied on to achieve compliance.

1. Glazing Basics

For the purposes of the BCA, glazing refers to windows, glazed doors and other transparent or translucent elements (such as glass bricks) located in the building fabric. It does not include roof lights. A glazing element includes the glass (or other glazing material), any air or gas fill, and the supporting frames. The performance of glazing is measured in terms of Total U-Value and Solar Heat Gain Coefficient.

It is essential that glazing elements are rated as a whole to account for the different thermal properties of the glass, any fills, the frame, and also the impact of the frame on the glass (edge-effect). These areas are indicated on the diagram to the right.

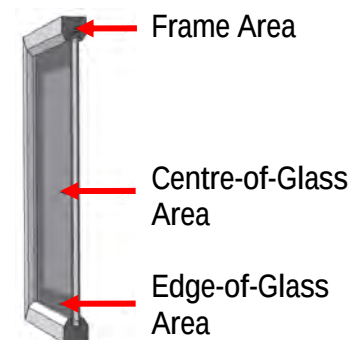


Figure courtesy of the Australian Window Association

The BCA requires glazing products to be rated in accordance with the [Australian Fenestration Rating Council](#) (AFRC) protocols and procedures. Standardised rating of glazing products ensures that a valid comparison can be made between the performance of different glazing products.

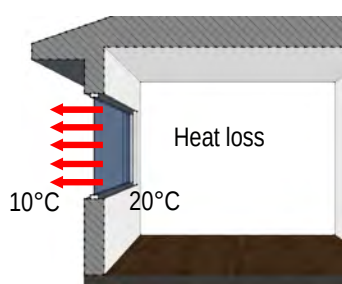
Glazing suppliers and manufacturers can provide the rated Total U-Value and SHGC for their glazing products.

1.1. Total U-Value

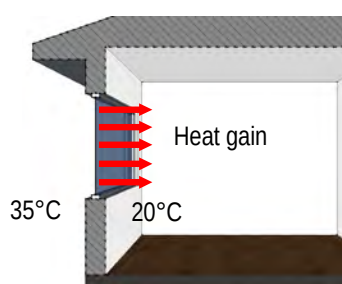
The Total U-Value is a measure of the ability of glazing (glass and frame together) to conduct heat. A lower U-value indicates a better insulator. Heat will flow from the warm side to the cool side as shown in the diagrams below.

1.2. Solar Heat Gain Coefficient (SHGC)

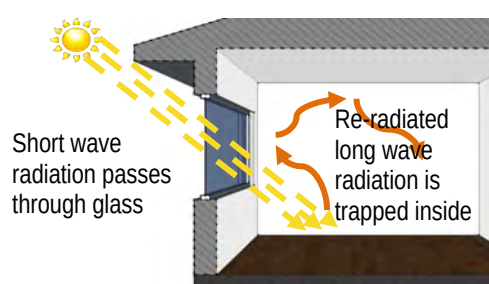
The SHGC is a measure of how readily glazing (glass and frame together) transmits radiant solar energy. It is the fraction of incident solar radiation that is admitted through the glazing, including both the amount directly transmitted and the amount absorbed and subsequently re-radiated inwards. A lower SHGC indicates that less of the solar energy falling on the glazing is admitted.



Conduction in winter



Conduction in summer



Solar Heat Gain

2. Changes in Calculation Methodology for Climate Zones 2 to 8

2.1. Conductance calculation

The conductance calculation for climate zones 2 to 8 no longer sets an allowance based on the floor area. The calculation is best thought of as “winter performance” and, despite being labelled “conductance”, is also affected by SHGC and shading. The “winter performance” aims to ensure that the total heat lost by conduction through the glazing will not exceed the direct solar gains available across all glazing in that storey of the dwelling. The new calculation has a number of implications:

- Lower U-values are generally better, as they limit the heat lost through the glazing in winter and the amount of unwanted heat gained by conduction during summer.
- In cooler climate zones, higher SHGCs are better, as they allow more direct solar gains.
- Using shading, rather than tinting, can allow compliance with the summer-based solar heat gain calculation while still achieving useful winter solar gains. Tinting (which lowers the SHGC) has the same effect in both summer and winter.
- Over-shading (that is, having too much shading) is heavily penalised in cooler climate zones as it limits the amount of direct solar gains in winter. This can be recognised when the calculator shows that the “% of allowance used” for solar heat gain is very low. Optimal shading for winter conditions still needs to be balanced with the solar heat gain, or “summer performance”.
- The winter solar gains are generally only high enough to be beneficial on the North, Northeast and Northwest orientations. The more glazing on the E-SE-S-SW-W orientations, the greater the amount needed to compensate on the NW-N-NE orientations. For alterations and additions, this can have a significant impact. Further guidance is provided below.
- Because the requirement does not depend on floor area, there is no arbitrary limit to the amount of glazing. Sometimes the answer is to put in more glazing (of a suitable type) on beneficial orientations. Changing the shape of glazing can also help. For an opening of a given area, lowering the sill can expose the glazing to winter sun while keeping it fully shaded in summer. Where it is not possible to add glazing on the beneficial orientations, it may be necessary to decrease the amount of glazing on the other orientations or to select glazing with lower U-values.

Note that the conductance calculation methodology for climate zone 1 has not changed.

2.2. Exposure factors

The calculation now uses exposure factors for both winter and summer performance. They depend on the orientation and the shading.

- E_w refers to the winter exposure factor. Higher numbers are better. Generally, a number greater than 1 indicates a “beneficial” exposure, that is, a good combination of orientation and shading for glazing in winter. Conduction compliance may be assisted by increasing glazing areas with “beneficial” exposures. Exposure factors below 1 can still be viable but may require glazing with better (that is, lower) U-values.
- E_s refers to the summer exposure factor. Lower numbers are better. A larger number indicates that, for the same glazing type and area, the glazing will use up more of the solar heat gain allowance.

Guidance on how to balance summer and winter glazing performance using the ratio of E_w to E_s (E_w / E_s) is given in Section 4.1. See also Section 3.6 and the calculator screenshot.

3. Changes to the Glazing Calculator Layout

The more significant additions and changes to the layout of the glazing calculator first issued with BCA 2010 (compared to previous versions) are identified below. The screenshot on the next page highlights these additions and changes.

3.1. Climate zone

The climate zone is now entered in the cell to the right of the building description.

3.2. Floor construction

The floor construction is now more clearly identified. Rather than the floor type being identified as Type A or B it is now described as “Direct Contact” or “Suspended”. Either or both types can be used. Where two floor types are nominated, they must be on the same storey. The calculator will determine a composite glazing stringency for the storey, taking account of the relative areas of each floor type. Previous calculators reported the outcomes separately for each floor type.

3.3. Orientation/facing sector

The orientation is now a single column due to the change in how the floor construction is entered.

3.4. Air movement

The air movement is now entered in the cell below the storey rather than the cell beneath the floor area, and is denoted by the words “Standard” or “High” rather than S or H. A number between 1 and 2 can be entered if the ventilation opening area is between Standard and High, and the solar heat gain allowance will be adjusted accordingly.

3.5. Wall insulation concession (input first added in BCA 2010 glazing calculator)

This cell refers to the optional trade-offs between wall insulation and glazing performance permitted by Table 3.12.1.3b for high-mass walls. Where such an insulation concession is used, the glazing stringency is increased as indicated in the top-right-hand corner of the glazing calculator.

3.6. Exposure factor (variable display first added in BCA 2010 glazing calculator)

Clicking in the title row under “Exposure” makes a drop-down menu available to change the exposure factor information currently displayed. Choose from either “ E_s ”, “ E_w ” or “ E_w / E_s ”. The first two are described in Section 2.2; E_w / E_s is described in Section 4.1.

Note that dwellings with multiple storeys still require a separate glazing calculator report for each storey.

4. Advanced Glazing Calculator Techniques

4.1. Balancing summer and winter performance using E_w / E_s

The E_w / E_s value indicates the relative impact of the winter and summer exposures for a given glazing orientation and shading arrangement. A number greater than 1 indicates a glazing element where there is significant benefit in winter without too much penalty in summer. A number lower than 1 indicates that increasing the area will cause a summer penalty greater than the winter benefit.

4.2. Element share of % of allowance used

This column appears twice: once for conductance ("winter performance") and once for solar heat gain ("summer performance"). For each of these two:

- The first percentage shows the relative contribution of each window. Changing a window with a higher element share % (for example, windows #1 and #11 in the screenshot below) will have much more impact than one with a lower element share % (eg. windows #5-8 below).
- The second percentage is the same for every element in that column, and shows how much of the total allowance is used by all the glazing combined. This must be no more than 100% to pass.

4.3. Device shading

One way of balancing winter and summer performance is to use a "device" (see 3.12.2.2(b) in BCA Volume Two) which can be adjusted to suit the season. Adjustable shading, such as a rollup awning, can be assessed in the calculator by completing two separate reports. A first report, for "summer performance", would include the "device" and demonstrate compliance with solar heat gain requirements by not exceeding 100% of the solar heat gain allowance. The second report, for "winter performance" would replace the "device" by shading P/H values which reflect its retracted state and demonstrate compliance for conductance by not exceeding 100% of the conductance allowance. The two reports, taken together, would provide evidence of overall compliance, provided they vary only in the extent of shading provided by the "device".

BCA VOLUME TWO GLAZING CALCULATOR (first issued with BCA 2010)

Building name/description: **House 1**

Storey: **Ground**

Air Movement: **Standard**

Climate zone: **6**

Floor Construction: **Direct contact**

Area: **159m²**

Area of storey: **159m²**

Area of glazing: **33.8m² (21% of area of storey)**

Number of rows preferred in table below: **12 (as currently displayed)**

CONSTANTS: C_u 6.418, C_{SHGC} 0.153

CONSTANT REDUCED BY: **15%**

ADJUSTED CONSTANT: **5.455**

ALLOWANCES: C_u **5.5**, $C_{SHGC} \times \text{Area}$ **24.3**

Wall insulation concession (requires increased glazing stringency)
Table 3.12.1.3b Climate zone 6 Option (a)/(i)

GLAZING ELEMENTS, ORIENTATION, SIZE and PERFORMANCE CHARACTERISTICS						SHADING		CALCULATION DATA		CALCULATED OUTCOMES - OK (if inputs are valid)				
Glazing element	Orientation	Size		Performance		P&H or device		Exposure	Size	Conductance - PASSED		Solar heat gain - PASSED		
ID	Description (optional)	Facing sector	Height (m)	Width (m)	Area (m ²)	Total U-Value (AFRC)	SHGC (AFRC)	P (m)	H (m)	Es	U x area / winter access	Element share of % of allowance used	SHGC x Es x area	Element share of % of allowance used
1	Living	N	2.10	4.20		3.30	0.68	0.55	2.20	8.82	1.40	26% of 98%	2.9	20% of 60%
2	Bed 2	N	2.10	2.10		3.30	0.68	0.55	2.20	4.41	0.70	13% of 98%	1.4	10% of 60%
3	Bed 1	E	2.10	1.80		3.30	0.68	0.55	2.20	2.72	0.60	11% of 98%	2.2	15% of 60%
4	Bed 3	E	2.10	1.80		3.30	0.68	0.55	2.20	2.72	0.60	11% of 98%	2.2	15% of 60%
5	Bath	S	0.60	0.60		3.30	0.68	0.55	0.70	0.06	1% of 98%	0.1	0.4% of 60%	
6	WC	S	0.60	0.60		3.30	0.68	0.55	0.70	0.06	1% of 98%	0.1	0.4% of 60%	
7	Laundry door light	S	0.60	0.60		3.30	0.68	0.55	0.70	0.06	1% of 98%	0.1	0.4% of 60%	
8	Laundry window	S	0.60	0.60		3.30	0.68	0.55	0.70	0.06	1% of 98%	0.1	0.4% of 60%	
9	Kitchen	S	0.90	2.40		3.30	0.68	0.55	1.00	2.16	0.34	6% of 98%	0.5	3% of 60%
10	Dining	S	2.10	1.80		3.30	0.68	0.55	2.20	3.78	0.60	11% of 98%	1.1	7% of 60%
11	Living	W	2.10	2.70		3.30	0.68	0.55	2.20	5.67	0.90	17% of 98%	4.1	28% of 60%

IMPORTANT NOTICE AND DISCLAIMER IN RESPECT OF THE GLAZING CALCULATOR

The Glazing Calculator has been developed by the ABCB to assist in developing a better understanding of glazing energy efficiency parameters. While the ABCB believes that the Glazing Calculator, if used correctly, will produce accurate results, it is provided "as is" and without any representation or warranty of any kind, including that it is fit for any purpose or of merchantable quality, or functions as intended or at all. Your use of the Glazing Calculator is entirely at your own risk and the ABCB accepts no liability of any kind.

If inputs (including air movement levels) are valid



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5. Glazing in Alterations and Additions

The information below is of a general nature as the application of the BCA provisions to alterations and additions is governed by State and Territory building legislation. The specific arrangements in the relevant State or Territory jurisdiction may be different to that described below. The final decision is at the discretion of the Building Control Authority and is dependent upon the particular characteristics of the existing building, and the type and extent of work being carried out. More information can be found in the *2010 Information Handbook for Energy Efficiency: BCA Volume Two* available from the [ABCB website](#).

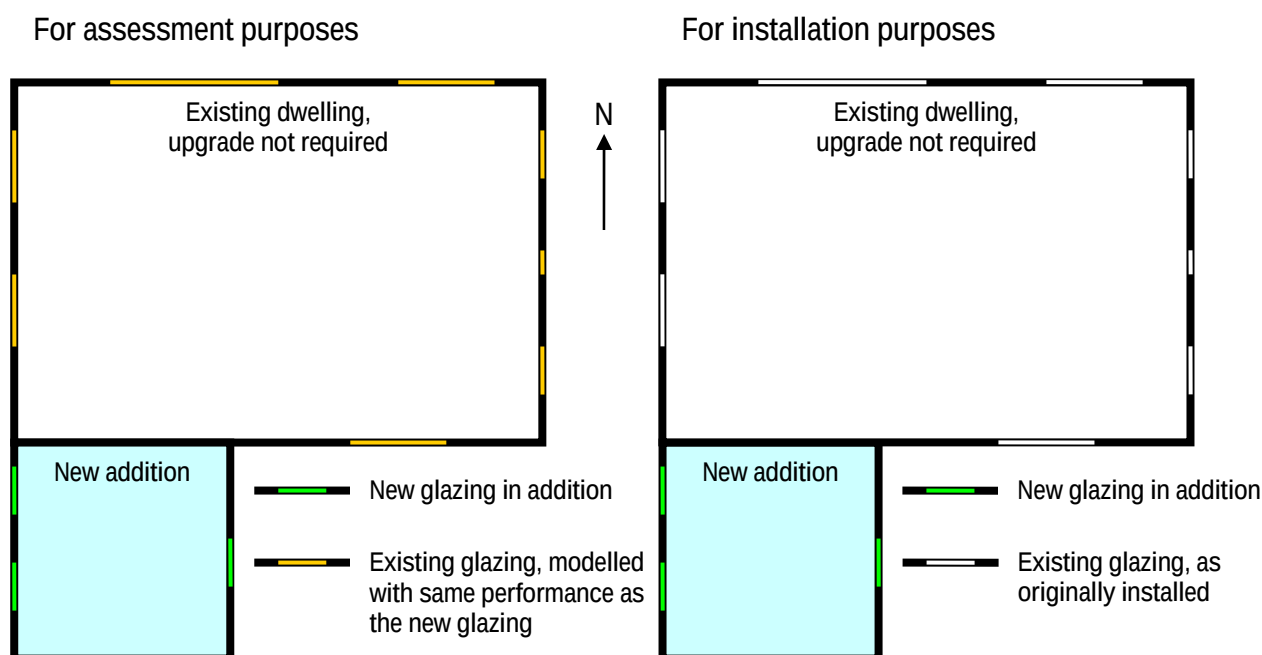
5.1. Where an upgrade is required by the State or Territory legislation

Where an upgrade to the existing dwelling is required, the existing glazing may also need to be replaced in accordance with requirements of the Building Control Authority.

5.2. Where an upgrade is not required by the State or Territory legislation

Where an upgrade to the existing dwelling is not required, it would be reasonable to assess the compliance as if the new glazing were uniformly applied to the whole storey, but to require the complying glazing to be *installed* only in the alteration or addition.

It may be very costly or even impossible for new glazing in an alteration or addition to compensate for the poor performance of existing glazing, especially for an alteration or addition that does not have any north-facing glazing in a cooler climate.



Remember to check with the Building Control Authority or the relevant State or Territory Administration as to whether or not this approach is applicable or acceptable in that jurisdiction.